REST API Interview Questions

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1. What is REST?

REST (Representational State Transfer) is an architectural style for designing networked applications. It uses HTTP methods to interact with resources.

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2. What is an API?

API (Application Programming Interface) allows applications to communicate and exchange data. REST APIs are one type of web API.

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3. Difference between REST and SOAP?

REST is lightweight, uses JSON/HTTP, and stateless. SOAP is protocol-heavy, uses XML, and has built-in security.

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4. What are HTTP methods used in REST?

Common methods: **GET** (read), **POST** (create), **PUT** (update/replace), **PATCH** (update/modify), **DELETE** (delete).

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5. What is statelessness in REST?

Each REST API request contains all necessary information. Server does not store client context between requests.

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6. What is a resource in REST?

A resource is any data object or entity exposed via REST API. Each resource has a unique URI.

7. What is a URI?

Uniform Resource Identifier (URI) uniquely identifies a resource. Example: `/users/123`.

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8. Difference between PUT and PATCH?

PUT replaces the entire resource. PATCH modifies only specified fields.

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9. What is the difference between POST and PUT?

POST creates a new resource and is not idempotent. PUT creates/updates a resource and is idempotent.

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10. What is idempotency?

An operation is idempotent if executing it multiple times has the same effect as executing it once. Example: GET, PUT, DELETE.

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11. What is HATEOAS?

HATEOAS (Hypermedia As The Engine Of Application State) allows REST clients to navigate API dynamically via links provided in responses.

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12. What are REST constraints?

REST constraints include **client-server, stateless, cacheable, layered system, uniform interface, code-on-demand (optional)**.

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13. What is the difference between PUT and POST for resource creation?

POST is used for creating resources where server assigns ID. PUT is used when client knows the resource ID.

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14. What is a REST endpoint?

A REST endpoint is a URI that clients use to access a resource. Example: `/users`, `/products/1`.

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15. What is versioning in REST API?

Versioning ensures backward compatibility. Example: `/v1/users` vs `/v2/users`.

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16. What is rate limiting?

Rate limiting restricts the number of API requests per time period to prevent abuse or server overload.

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17. What is API key authentication?

API key is a token passed with requests to identify and authorize the client.

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18. What is JWT?

JWT (JSON Web Token) is a compact token used for authentication and data exchange between client and server.

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19. Difference between REST and GraphQL?

REST exposes multiple endpoints per resource. GraphQL exposes single endpoint allowing clients to request exactly the data they need.

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20. What is content negotiation?

Content negotiation allows server and client to agree on response format (JSON, XML, HTML) via `Accept` headers.

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21. What are common HTTP status codes?

- 200 OK (success)
- 201 Created (resource created)
- 204 No Content (successful with no body)
- 400 Bad Request (client error)
- 401 Unauthorized
- 403 Forbidden
- 404 Not Found
- 500 Internal Server Error

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22. Difference between 200, 201, and 204?

200 = request successful with response body
201 = resource created successfully
204 = request successful, no content returned

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23. Difference between 4xx and 5xx errors?

4xx = client-side error, 5xx = server-side error

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24. Difference between REST API and WebSocket?

REST is request-response, stateless. WebSocket provides bi-directional, persistent connection for real-time data.

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25. What is CORS?

Cross-Origin Resource Sharing (CORS) allows browsers to make requests to a different domain safely.

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26. Difference between SOAP and REST?

Already explained — REST is lightweight, uses HTTP and JSON; SOAP is protocol-heavy, uses XML.

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27. What is a RESTful API?

A RESTful API follows REST constraints: stateless, cacheable, layered, uniform interface, and uses HTTP methods.

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28. Difference between HTTP and HTTPS?

HTTPS encrypts data using SSL/TLS. HTTP sends data in plain text.

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29. What is caching in REST API?

Caching stores response data temporarily to reduce server load and improve performance. Controlled using headers like `Cache-Control`.

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30. What is the difference between PUT and POST idempotency?

PUT is idempotent (repeated calls same effect). POST is non-idempotent (creates multiple resources if repeated).

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31. What is rate limiting?

Already explained — limits number of API calls per time frame to avoid overload.

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32. Difference between XML and JSON?

JSON is lightweight, easy to parse, and preferred in REST. XML is verbose and requires parsing libraries.

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33. What is OpenAPI (Swagger)?

OpenAPI (Swagger) is a specification for documenting REST APIs. It allows auto-generating docs and client SDKs.

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34. What is the difference between PUT, POST, PATCH, DELETE, GET?

- GET = read
- POST = create

- PUT = replace
- PATCH = partial update
- DELETE = remove

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35. Difference between query parameters and path parameters?

Query parameters appear after `?` and are optional. Path parameters are part of URI and identify resource.

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36. What is a client-server architecture in REST?

Client and server are separate. Client requests resources; server provides them. Separation allows independent evolution.

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37. What is stateless vs stateful?

Stateless = server does not store client info between requests. Stateful = server maintains session info.

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38. Difference between JSON and XML?

JSON = lightweight, readable, preferred. XML = verbose, needs parsing, supports schema.

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39. What is API throttling?

Throttling limits number of requests a client can make in a time window to prevent server overload.

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40. Difference between synchronous and asynchronous API calls?

Synchronous = client waits for server response
Asynchronous = client continues execution, handles response later

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41. What is response body?

Response body contains data sent back by server after API request. Usually JSON in REST APIs.

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42. What is request header?

Request headers provide metadata about the request, such as content type, authorization, and user agent.

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43. Difference between API authentication and authorization?

Authentication = verify identity of client (who you are)
Authorization = verify permissions (what you can do)

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44. What is OAuth 2.0?

OAuth 2.0 is an authorization framework that allows secure access to resources without sharing credentials.

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45. Difference between JWT and OAuth?

JWT is a token format for authentication. OAuth 2.0 is a protocol for granting limited access using tokens.

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46. What is the difference between PUT and PATCH?

Already explained — PUT = full replacement, PATCH = partial update.

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47. Difference between REST and gRPC?

REST = HTTP/JSON, text-based, stateless
gRPC = HTTP/2, binary, supports streaming, faster

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48. What is API versioning and why is it important?

API versioning ensures backward compatibility when API changes. Example: `/v1/users` vs `/v2/users`.

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49. What is rate limiting vs throttling?

Rate limiting = max requests per time period
Throttling = slows down or blocks requests beyond limit

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50. What are best practices for REST APIs?

- Use nouns in endpoints
- Use proper HTTP methods
- Stateless design
- Versioning
- Proper status codes
- Security (HTTPS, authentication)